EPFL

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THESIS DISCUSSION · PROF. PRENDINGER

What should M2 learn to call a good trade?

Triple-barrier labels across 20 assets:
volatility, direction, and a controlled target experiment.

SCOPE	SecondaryModel's own 20 assets · daily candles · 7-day maximum holding period
EVIDENCE	Training diagnostics · verified M1 baseline · Pablo's saved M2 reference
PREPARED	22 October 2026 · short discussion for next week
UPDATE	29 October 2026 · UP M1 Sharpe −2.5744 · 48 / 48 fits reviewed

Which dates belong to which results?

Configured cutoffs · daily horizon = 7 bars · all dates below use YYYY-MM-DD.

Configuration	Data start	Train ends	Validation ends	Data ends
Earlier trial · separate PDF	2024-07-01	2025-05-30	2025-10-01	2026-01-25
Pablo GPU8 · now copied	2024-07-01	2025-08-31	2025-12-31	2026-08-01

Actual sample dates · after context and split rules

New daily inputs · Pablo's configuration
Train: 2024-06-30 – 2025-08-31
Val-Cal: 2025-09-08 – 2025-10-18
Val-Opt: 2025-10-26 – 2025-12-31
Test: 2026-01-08 – 2026-07-18

Other evidence and historical context
Label diagnostics: 2024-07-01 – 2025-08-24
Historical AG / Kumo Test · separate PDF:
2025-10-09 – 2026-01-18
Separate LightGBM Test:
2026-01-01 – 2026-09-23

The original loader creates a first training window on **2024-06-30**, one day before the configured start. Its context and embargo conventions are preserved. The realised Test dates match Pablo's saved reports.

GPU8 config observed 2026-10-27; daily input checks 2026-10-28; saved earlier trial, diagnostics and benchmark dates. Full LightGBM train/validation ranges accompany its result slides.

VERIFIED ON 28 OCTOBER 2026 · M1 BASELINE

The reproduced UP baseline has Sharpe −2.5744

Test: 2026-01-08 – 2026-07-18 · spread equity · 7-day horizon · fee 0.2%.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−54.25%	**−2.5744**	−57.99%	1045
DOWN	30.44%	1.7423	−17.04%	2580

12 / 12

Target / direction baseline checks passed

27

Weekly return observations from 192 daily bars

−39.86%

Buy-and-hold return on the same Test period

Buy-and-hold also lost money in this period. All six targets share M1's Test trades. **Four-backend comparison: 48 / 48 fits reviewed.** AutoGluon, Kumo, TabPFN and Random Forest are all complete.

Independent daily data preparation reproduces saved GPU8 M1 metrics within 0.0001; existing UP multigranularity cache also checked. Spread equity smooths trade returns; Sharpe uses ×√(365.25/7). The inspected Test is exploratory.

Return and drawdown explain the “−45%” figure

Same 1045 UP trades · 2026-01-08 – 2026-07-18 · different equity calculations.

Equity calculation	Total return	Max drawdown	Sharpe
Spread · original baseline	−54.25%	−57.99%	−2.57
Cash · open positions at cost	−43.60%	−45.35%	−2.79
Cash · current market prices	−43.66%	−48.75%	−2.14

Saved cash report

−43.60% is the endpoint loss.
−45.35% is the worst fall from a prior peak.
Open positions remain at entry value.

Current cash code

Open positions follow current closing prices.
Interim losses become visible: **−48.75% drawdown**.
Use the same valuation rule for comparisons.

Replay of Pablo’s saved UP trade CSV on 2026-10-28; cash exposure 0.10 in both rows. Saved cash metrics match positions held at cost. Final-day entries can remain open: cash figures are endpoint equity, including the current marking fee.

Pablo’s saved UP M2 Sharpes are also negative

Test entries: 2026-01-08 – 2026-07-18 · 1d · TP labels · spread-equity engine.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−54.25%	**−2.57**	30.44%	1.74
AutoGluon	−54.22%	**−2.36**	22.36%	1.58
CTTS	−58.65%	**−2.23**	23.36%	1.45
TabPFN	−21.50%	**−1.07**	20.70%	1.70
TabICL	−57.91%	**−2.43**	35.95%	1.91

These are **Pablo’s saved TP results**. The completed 48-fit matrix uses **fresh TP controls for each backend**. M1, dates and the backtest match; the new M2 scores differ from these saved classifiers.

Saved GPU8 outputs fetched 2026-10-27; 48 replayed metrics agree within 0.0001. New models are separate fits on Izar, with pinned principal packages; hardware, Python patch release and fitted ensembles can differ. Exact M2 reproduction is not established.

Inspecting the labels does not require retraining M2

Past candles
Market features + frozen M1
M1 proposes UP or DOWN.

→

Observed future candles
Apply the chosen exit rule.
Compute the net outcome.

→

Meta-label z
1 if the proposal paid;
0 otherwise.

WHAT WE CAN SHOW NOW

No retraining

Compare volatility, barrier touches and label changes using the same historical M1 calls.

WHAT A PERFORMANCE CLAIM NEEDS

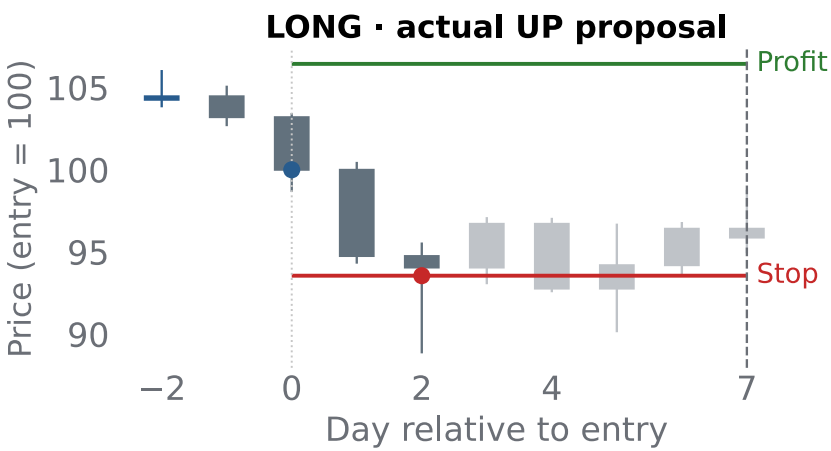
Retrain the secondary model

To test whether M2 learns the new target better, fit it on the new labels. Reuse M1's forecasts.

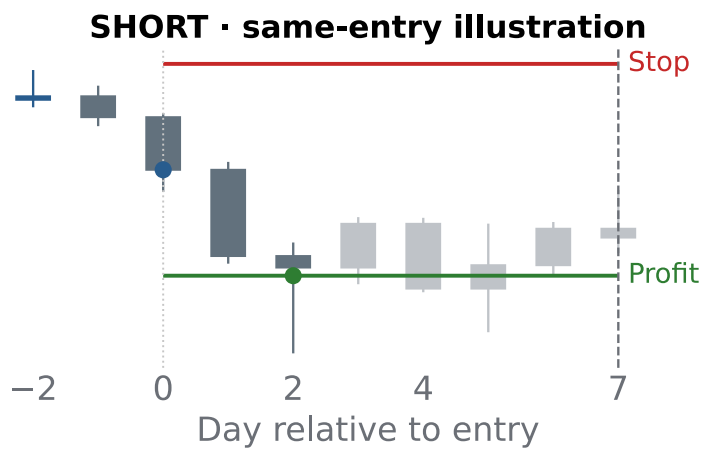
AFML §§3.4–3.7. Entry dates: 2024-07-01 – 2025-08-24. 7,593 eligible UP/DOWN proposals from 19 of SecondaryModel's 20 assets; TON unavailable in this parquet.

The same price move can stop a long and profit a short

BTC · 2024-07-03 · entry at day-0 close · symmetric k = 1 barriers.



Exit day 2 · net -6.6% · meta-label 0



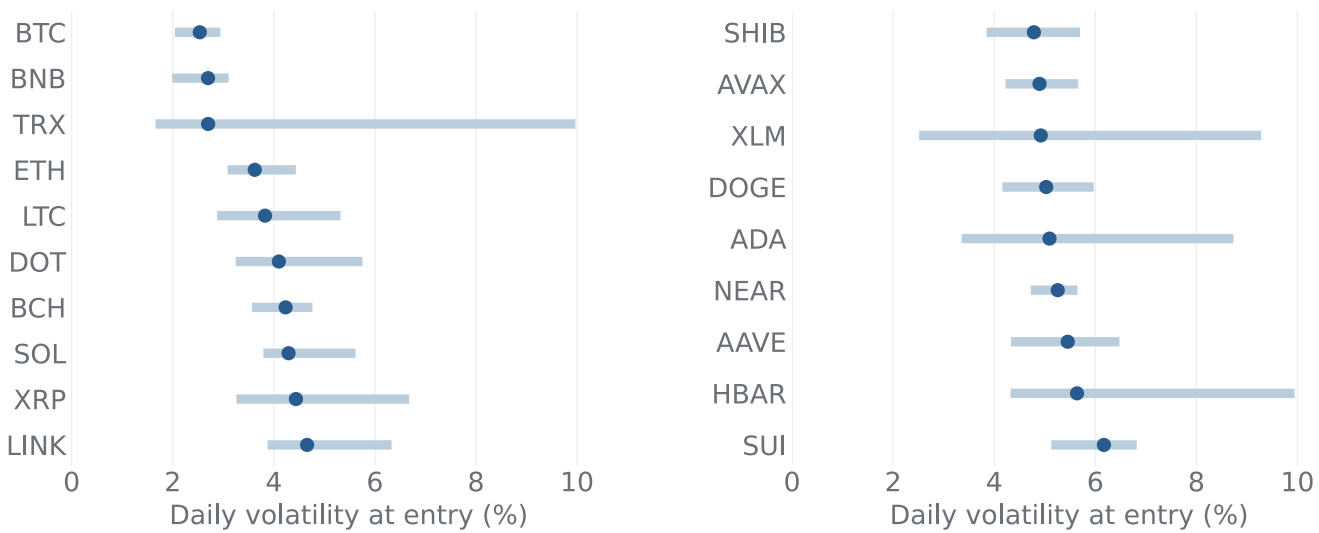
Exit day 2 · net +6.2% · meta-label 1

UP / long: profit above entry, stop below. DOWN / short: profit below, stop above.
The same entry-time volatility is used for both directions.

Actual training-period candles; explanatory episode selected for a stop. SHORT is a counterfactual, not a second Kronos call. Cost 0.2%; assumed barrier fills.

One common k still gives a different width at each entry

$$w_{a,t} = k\sigma_{a,t}\sqrt{7}$$

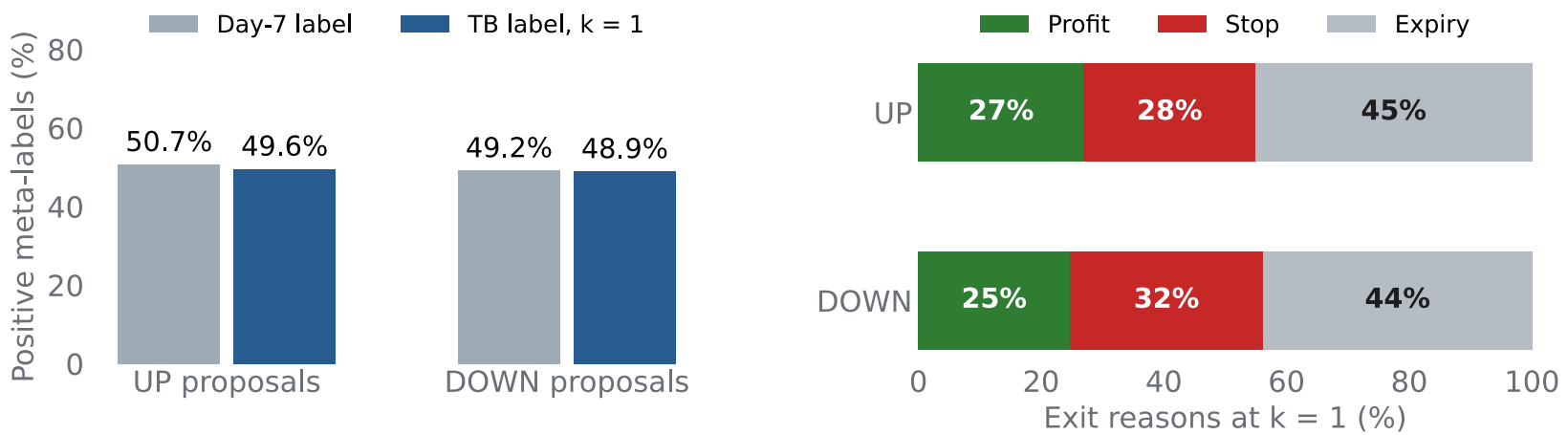


At k = 1, median widths are **6.7% for BTC**, **9.6% for ETH**, and **16.3% for SUI**.
σ changes with the asset and date; the chosen barrier is then fixed for that event.

Entry dates: 2024-07-01 – 2025-08-24. EWMA(100), known at entry; dots = median, lines = entry-date 10th–90th percentiles; √7 scaling.

Apply TBM to both sides; report the outcomes separately

1,921 UP proposals · 5,672 DOWN proposals · same k = 1 and same volatility estimator.

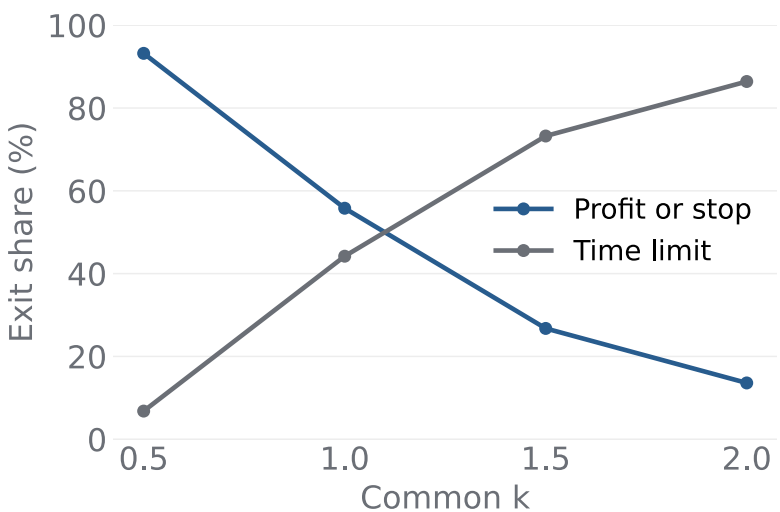


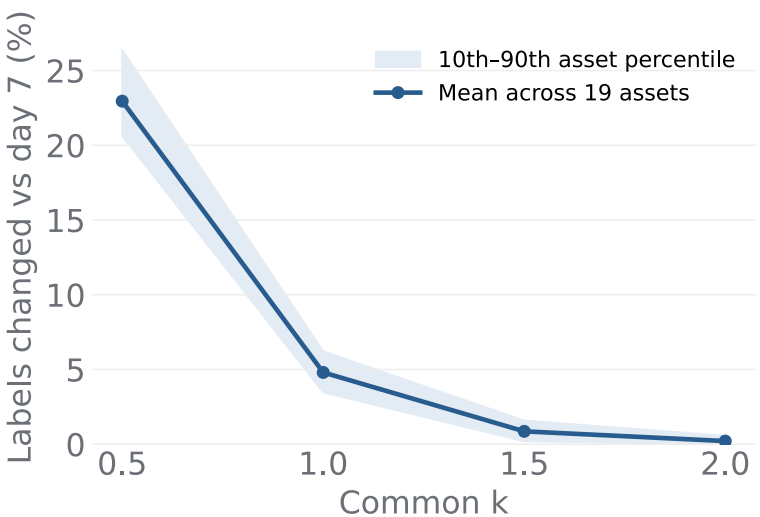
Meta-labels change for **5.9% of UP calls** and **4.4% of DOWN calls**.
These groups contain different dates and market conditions; this is a descriptive comparison.

Entry dates: 2024-07-01 – 2025-08-24. 19 of 20 assets; high/low touches, same-bar double touch = stop; cost 0.2%. Positive-label share is not M2's predictive accuracy.

A similar binary label can hide a different trade

Small k: closer exits. Large k: more trades remain open until day 7.





At k = 1: **55.8% horizontal exits**, but only **4.8% changed labels**, averaged across assets. The exit, return and holding time can change while “profitable / unprofitable” stays the same.

Entry dates: 2024-07-01 – 2025-08-24. Equal asset weights. Shading: 10th–90th percentiles across assets. Training diagnostics; these charts do not select an optimal k.

Start with a global k; select it for the intended trade

Start with one shared parameter

- Same k for all 20 assets and both sides.
- Volatility already adapts the width.
- Small grid: 0.5, 1, 1.5, 2.
- Keep the horizon, features and cost model fixed.

Choose on validation data

- Specify the intended exit policy first.
- Rank candidates by a predefined net portfolio objective.
- Check drawdown, coverage and stability across assets and time.
- Confirm the choice on fresh data.

There is no universal “ideal k”. Add separate direction or asset parameters only if they bring a stable validation benefit. A 50/50 label balance is not an optimization objective.

Proposed research protocol. AFML ch. 7 and ch. 11; López de Prado’s backtest-overfitting work. Existing 2026 test results have already been inspected.

READING THE COMPARISON

TP means true-positive meta-labeling

Original TP asks whether a Kronos proposal made money at day 7, **after the 0.2% fee**.

Label 1 · true positive

The proposed UP or DOWN trade was profitable after the fee at day 7.

Label 0 · unsuccessful proposal

Its signed day-7 return was zero or negative after the fee.

Column	M2 training target	Purpose
TP	Original seven-day true-positive label	Original target baseline for each backend
TB ref	Same TP label; TB eligibility filter	Control with matched training eligibility
TB k	Profit / stop / time-barrier outcome	Test the effect of a different target

TP is not a take-profit order. M2 predicts success from the available features; it does not know the future label. All reported backtests still execute at day 7, including models trained on TB targets.

Original conversion: pred=1 and lab=1 → meta_label=1; pred=1 and lab=0 → 0. The signed day-7 rule agrees with all 18,230 complete-horizon proposals in the 40 daily CSVs. RF = Random Forest; TabPFN is the original GPU8 backend.

Four backends, six targets: 48 / 48 fits reviewed

Test: 2026-01-08 – 2026-07-18 · TP = true positive · fixed day-7 execution.
M1 Sharpe: **UP −2.5744** · DOWN 1.7423. RF = Random Forest.

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−2.199	−2.375*	−2.246	−2.376*	−2.305	−2.280
AutoGluon · DOWN	0.339	1.452	1.822	1.357	1.383	1.080
Kumo · UP	−1.983	−2.255	−2.463	−2.714	−2.038	−2.225
Kumo · DOWN	1.884	2.001	−0.278	1.731	1.849	1.730
TabPFN · UP	−2.527	−2.570	−2.353*	−2.594	−2.445	−2.519
TabPFN · DOWN	0.882	0.988	1.569	1.046	0.829	0.797
RF · UP	−1.800	−1.517	−2.222	−1.779	−1.388	−1.810
RF · DOWN	1.201	1.246	−0.473	−0.256	1.222	0.825

All UP scores remain negative. RF at k = 1.5 has the least-negative UP Sharpe, **−1.388**. The highest observed DOWN score is **Kumo TB ref: 2.001**, retaining TP labels.

48 fits; 288 return/drawdown/Sharpe metrics replayed within 0.0001. Annualized spread equity, 27 horizon returns. * = utility constraint failed, fallback threshold 0.5. TabPFN/RF use frozen GPU8 tuning parameters. Exploratory Test.

How much does Sharpe change versus the fresh TP fit?

Δ Sharpe = target Sharpe − TP Sharpe · Test 2026-01-08 – 2026-07-18.

M2 fit	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−0.176	−0.047	−0.178	−0.106	−0.081
AutoGluon · DOWN	+1.114	+1.483	+1.019	+1.044	+0.742
Kumo · UP	−0.272	−0.480	−0.731	−0.055	−0.242
Kumo · DOWN	+0.117	−2.162	−0.153	−0.035	−0.154
TabPFN · UP	−0.043	+0.174	−0.067	+0.082	+0.008
TabPFN · DOWN	+0.106	+0.687	+0.164	−0.054	−0.085
RF · UP	+0.283	−0.422	+0.021	+0.412	−0.010
RF · DOWN	+0.045	−1.675	−1.458	+0.020	−0.377

At k = 0.5, DOWN gains are **AutoGluon +1.483** and **TabPFN +0.687** versus TP. RF UP at k = 1.5 improves by **+0.412**, but its Sharpe remains negative.

This contrast includes the eligibility change: TB ref and TB-k remove **13 UP / 64 DOWN** early TON training proposals. The matched comparison follows.

Same realized Test trades, fees, features and fixed day-7 execution. Fresh TP fits differ from Pablo's saved M2 fits. Deltas are descriptive; this inspected Test does not select an optimal k.

DELTA 2 · MATCHED TRAINING ELIGIBILITY

TB label effects depend on backend and direction

Δ Sharpe = TB-k Sharpe − TB-ref Sharpe · TB ref keeps TP labels on the TB-eligible samples.

M2 fit	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	+0.128	−0.002	+0.069	+0.094
AutoGluon · DOWN	+0.370	−0.095	−0.069	−0.372
Kumo · UP	−0.208	−0.459	+0.217	+0.030
Kumo · DOWN	−2.279	−0.270	−0.152	−0.271
TabPFN · UP	+0.217	−0.024	+0.125	+0.051
TabPFN · DOWN	+0.580	+0.058	−0.160	−0.191
RF · UP	−0.704	−0.261	+0.130	−0.293
RF · DOWN	−1.719	−1.502	−0.024	−0.421

TabPFN · DOWN · k = 0.5

Sharpe **1.569**: **+0.580** versus TB ref.
Return **+34.78%**; max drawdown **−17.51%**.
Still below M1's DOWN Sharpe of 1.742.

Random Forest · UP · k = 1.5

Sharpe **−1.388**: **+0.130** versus TB ref.
Return **−34.61%**: improvement still loses money.
RF DOWN k = 1.5 trails TB ref by 0.024.

All TB variants share training eligibility and Test trades. This is a label comparison with day-7 execution, not early barrier-exit trading. One fit per cell; 27 horizon returns; no significance or optimal-k claim.